

```
import numpy as np
import pandas as pd
from sklearn.preprocessing import MinMaxScaler, StandardScaler
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("C:/Users/sreev/Downloads/diabetes.csv")
df.head()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
0	6	148	72	35	0	33.6
1	1	85	66	29	0	26.6
2	8	183	64	0	0	23.3
3	1	89	66	23	94	28.1
4	0	137	40	35	168	43.1

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

```
df.info()
print(df.isnull().sum())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Pregnancies                          768 non-null    int64
1   Glucose                              768 non-null    int64
2   BloodPressure                        768 non-null    int64
3   SkinThickness                        768 non-null    int64
4   Insulin                              768 non-null    int64
5   BMI                                  768 non-null    float64
6   DiabetesPedigreeFunction             768 non-null    float64
7   Age                                  768 non-null    int64
8   Outcome                              768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
Pregnancies      0
Glucose           0
BloodPressure     0
```

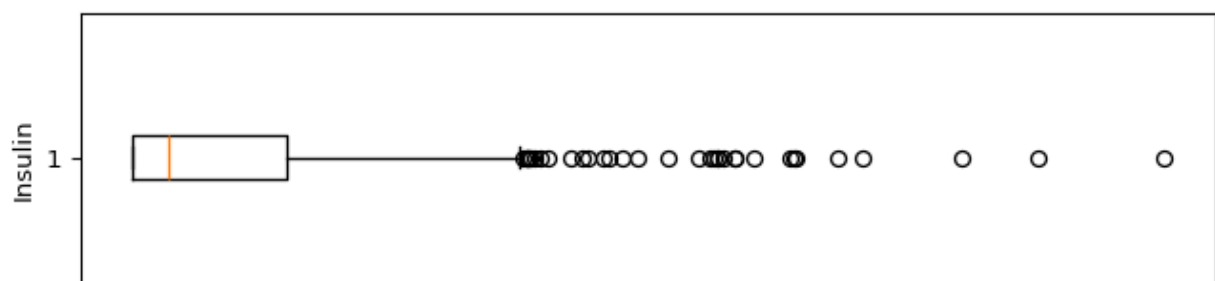
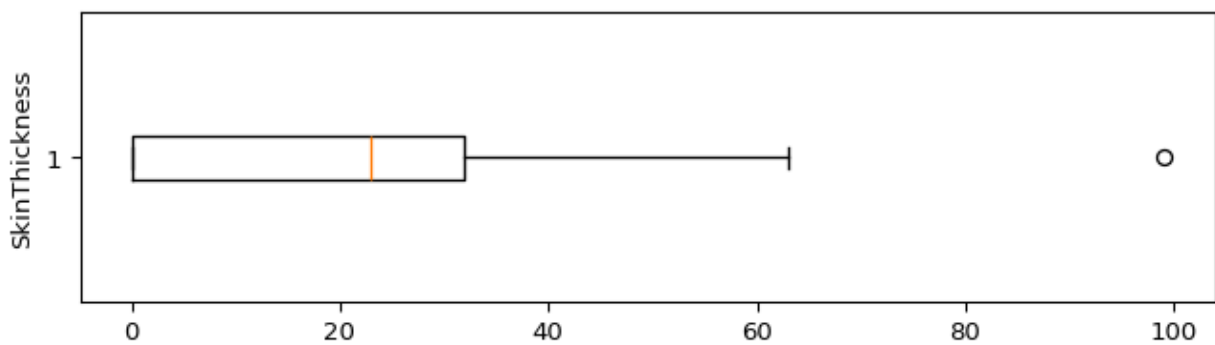
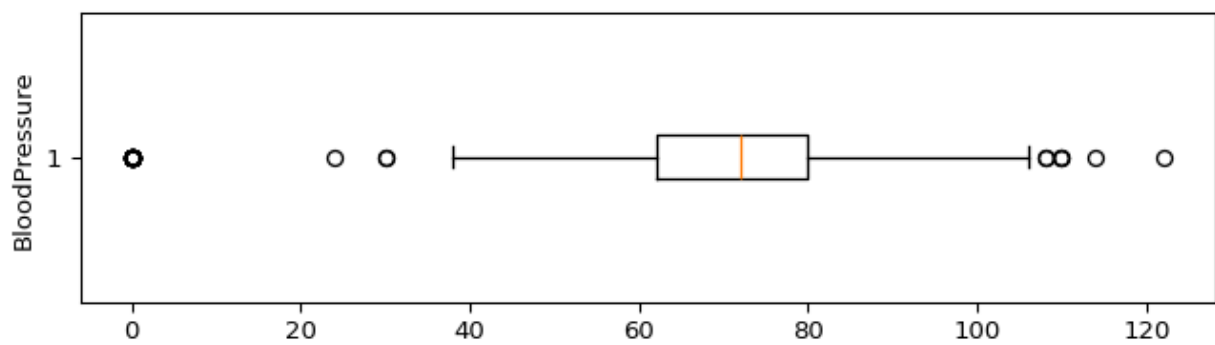
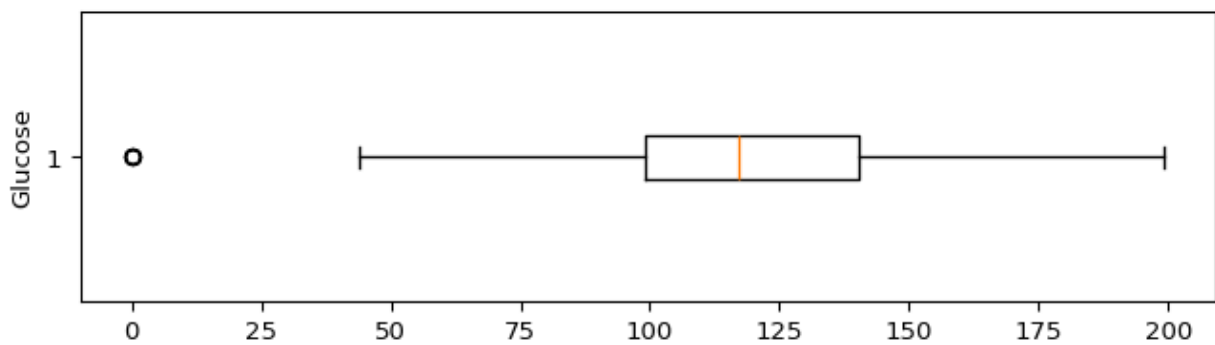
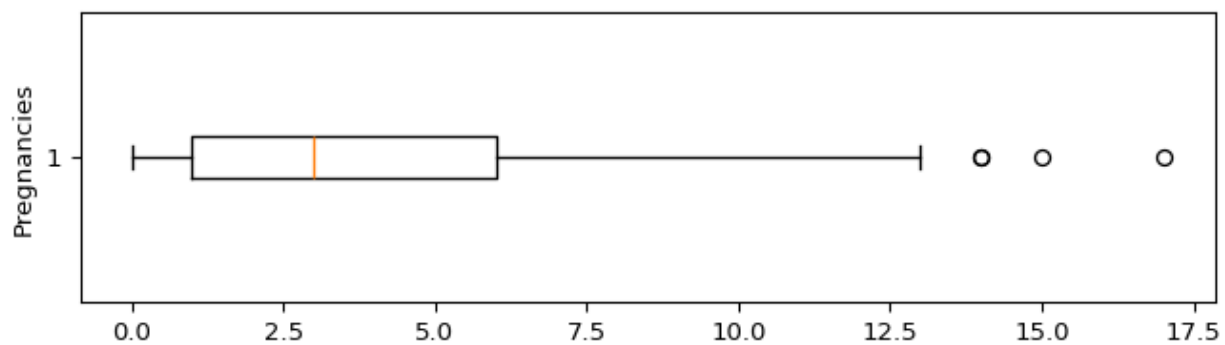
```
SkinThickness      0
Insulin            0
BMI               0
DiabetesPedigreeFunction  0
Age               0
Outcome            0
dtype: int64

df.describe()

fig, axs = plt.subplots(len(df.columns), 1, figsize=(7, 18), dpi=95)

for i, col in enumerate(df.columns):
    axs[i].boxplot(df[col], vert=False)
    axs[i].set_ylabel(col)

plt.tight_layout()
plt.show()
```



```

q1, q3 = np.percentile(df['Insulin'], [25, 75])
iqr = q3 - q1

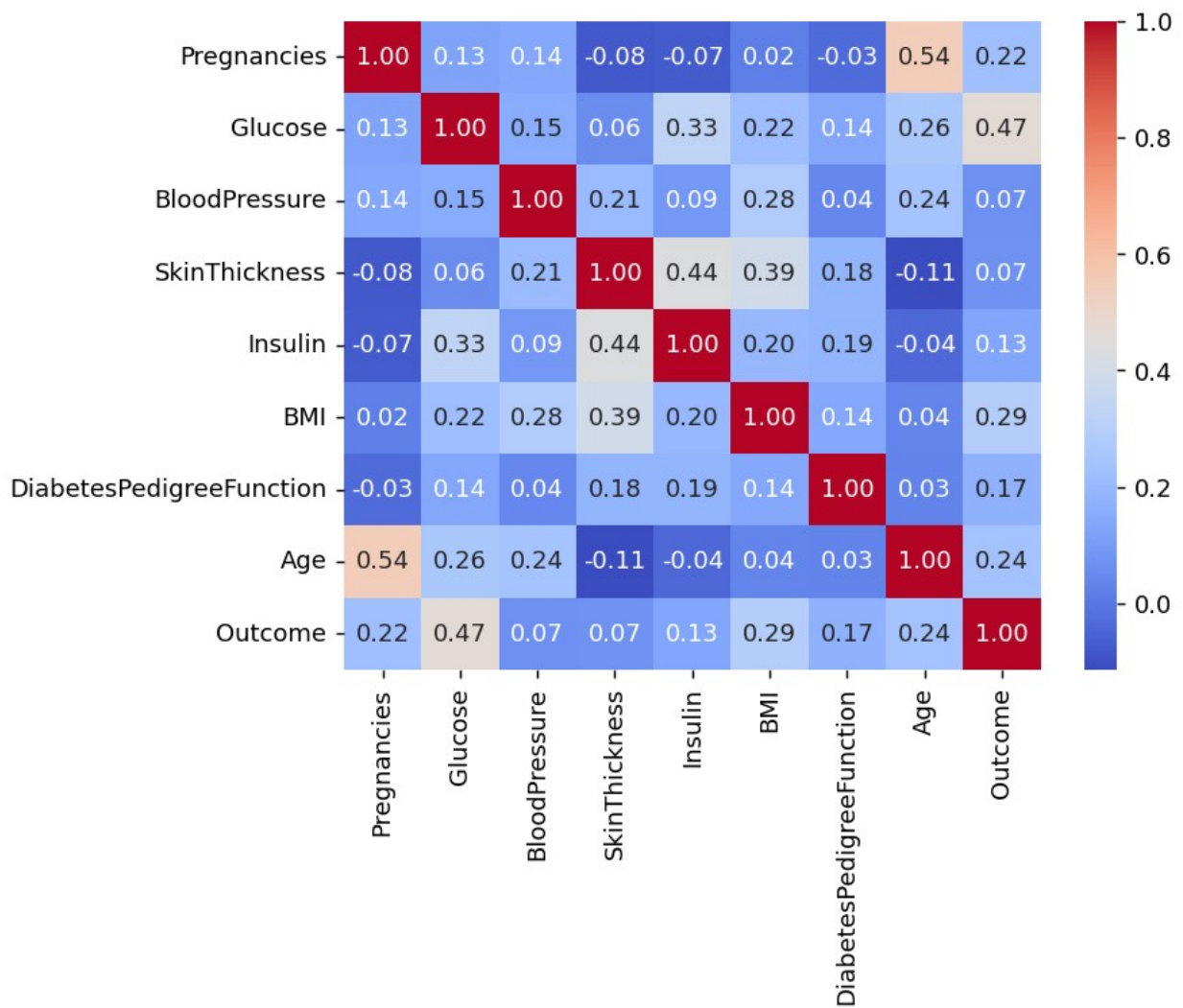
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr

clean_df = df[(df['Insulin'] >= lower) & (df['Insulin'] <= upper)]

corr = df.corr()

plt.figure(dpi=130)
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm')
plt.show()
print(corr['Outcome'].sort_values(ascending=False))

```



```

Outcome      1.000000
Glucose      0.466581

```

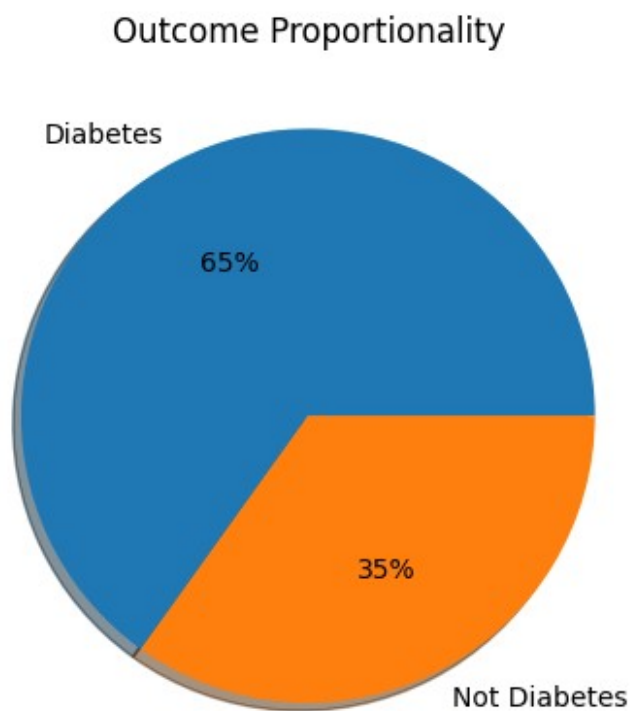
```

BMI                0.292695
Age                0.238356
Pregnancies        0.221898
DiabetesPedigreeFunction  0.173844
Insulin            0.130548
SkinThickness      0.074752
BloodPressure      0.065068
Name: Outcome, dtype: float64

plt.pie(
    df['Outcome'].value_counts(),
    labels=['Diabetes', 'Not Diabetes'],
    autopct='%.f%%',
    shadow=True
)

plt.title('Outcome Proportionality')
plt.show()

```



```

X = df.drop(columns=['Outcome'])
y = df['Outcome']

from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()

```

```

X_normalized = scaler.fit_transform(X)

print(X_normalized[:5])

[[0.35294118 0.74371859 0.59016393 0.35353535 0.          0.50074516
  0.23441503 0.48333333]
 [0.05882353 0.42713568 0.54098361 0.29292929 0.          0.39642325
  0.11656704 0.16666667]
 [0.47058824 0.91959799 0.52459016 0.          0.          0.34724292
  0.25362938 0.18333333]
 [0.05882353 0.44723618 0.54098361 0.23232323 0.11111111 0.41877794
  0.03800171 0.          ]
 [0.          0.68844221 0.32786885 0.35353535 0.19858156 0.64232489
  0.94363792 0.2          ]]

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```